Activity E1 (1 point)

Name and surnames:

Answer the following questions always reasoning the answers. Concision and accuracy in responses are valued.

1. What is the *high scalability* of the Worldwide LHC (Large Hadron Collider) Computing Grid based on?

2. Explain with an example how adding a property to a distributed system is often in detriment to other properties or system performance.

3. Which namespace is more *scalable*, the one provided by IPv6 or the one based on URLs? Why?

4. *An SMS messaging system does not provide transparency in the location, while* Gmail *does.* Why? (Note: consider that in an application such as these the "resource" to "locate" is the recipient user of the message).

5. *It is more complex to manage consistency in a replicated system than in one with a single server.* Is this statement true or false? Why?

6. Give a reason that explains why *session semantics* is adequate in a cloud file system such as Dropbox (and the one we are building in the course project).

7. We have a service supported with two PCs, named "primary" and "secondary" respectively, and a *log* file on a disk shared by both computers (in a local area network) that collects all the history of updates. When the primary PC fails, we manually update the secondary PC from the *log* and continue the service with it. Does this system provide greater *reliability* than one with a single PC?

8. Does the service of the previous exercise provide *transparent fault tolerance*?

9. How does an application like Dropbox preserve *user portability*?

10. Consider the following two statements: (a) *The development of the TCP/IP communications protocol proved decisive for the expansion of the Internet*. (b) *The widespread adoption of TCP/IP as a communications protocol proved decisive for the expansion of the Internet*. Which of these two statements do you think reflects more adequately the historical relevance of this protocol?